

Applied Mathematical Methods for Computing

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Module 1

Algebra, vectors and matrices

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Module 2

Sequence and Series

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Module 3

Combinatorics

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Lesson 3.1 Counting

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Introduction

Video • 1 min
- ✓

Counting

Video • 5 min
- ✓

Complex counting

Video • 4 min
- ✓

Counting

Practice Assignment • Grade: 100%
- ✓

Inclusion–Exclusion Principle

Dialogue • 30 min

3.2 The Pigeonhole Principle

- ✓

The Pigeonhole Principle

Video • 3 min
- ✓

The Pigeonhole Principle – examples part 1

Video • 4 min
- ✓

The Pigeonhole Principle – examples part 2

Video • 3 min
- ✓

Use the Pigeonhole Principle

Practice Assignment • Grade: 100%
- Think of an example

Discussion Prompt • 20 min
- ✓

Exercises with hints and tips

Reading • 1h

Lesson 3.3 Order Matters

- ✓

Permutations

Video • 6 min
- ✓

Permutations - solving a case study of your choice

Practice Assignment • Grade: 100%
- ✓

Permutations

Practice Assignment • Grade: 100%
- ✓

Combinations

Video • 10 min
- ✓

No order/combinations

Practice Assignment • Grade: 100%
- ✓

Combinatorics

Practice Assignment • Grade: 100%

3.4 Summary and Assessments

- ✓

Exercises with hints and tips

Reading • 10 min
- Conclusion

Video • 3 min
- Check your understanding: End of module 3

Graded Assignment • 20 min

Module 4

Probability and Statistics

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Lesson 4.1 Probability Definition and Properties

- ✓

Introduction to probability

Video • 11 min
- Probability

Practice Assignment • 30 min

Lesson 4.2 Statistics

- Introducing statistics

Video • 10 min
- Worked examples (socks from a drawer)

Video • 7 min
- Sharing birthdays

Exercises with hints and tips

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0:10 / 2:11

2x





Now it’s time to put into practice the concepts that you have learnt so far in Week 6. Attempt the following exercises. If you get stuck, you can refer to the hints and tips at the bottom of the page.

Please note that these exercises are optional, further practice – but we strongly recommend that you engage with them in order to test your knowledge and see where you might need to do additional study.

1.

A zip code in Canada consists of three letters and three digits. Each zip code begins with a letter. The letters and digits alternate; for instance, A1B2C3. Find the number of zip codes that:

a.

end in 6

b.

begin with A and end in 3

c.

end in Z

d.

are possible.
2.

Find the number of bytes that:

a.

begin with 101

b.

end with 110

c.

begin with and end in the same bit

d.

begin with and end in different bits

e.

have the same third and fourth bits.
3.

The password for a computer system consists of eight distinct alphabetic characters. Find the number of passwords possible that:

a.

end in the string MATH

b.

begin with the string CREAM

c.

contain the word COMPUTER as a substring

d.

contain the letters WORD together, but in any order.
4.

Find the number of three-digit numerals that can be formed using the digits 2, 3, 5, 6 and 9 if repetitions are not allowed.

Hints and tips for each question:

1.

Consider 6 places. The first place should always be a letter, the second should be a digit, and so on. Based on the given constraints, calculate each case. For example, in part a, we have $26 \cdot 10 \cdot 26 \cdot 10 \cdot 26 \cdot 1$. We are assuming digits and letters both can be repeated.
2.

Each byte is 8 bits, so we have 8 places and each place can be either 0 or 1. Calculate the possible number of bytes based on the given constraints. For example, in part a, it is $1 \cdot 1 \cdot 1 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ as the first 3 bits should always be 101.
3.

In this question we have 8 places, each of which can be any letter. So for each place we have 26 different possible values. Now, calculate each part based on the given constraints.
4.

We have 3 places with 5 different values (non-repetitive).

Go to next item

✓ Completed

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